

Technology Stack (Architecture & Stack)

Auto Ticket Classification using Flow Designer

Date	2 oct 2026
Team ID	SWTID-2026-5846
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

1. Technical Architecture

The application is implemented as a ServiceNow cloud-based configuration. The user interacts with a ServiceNow ticket form. The custom Incident WorkFlow table stores ticket information. Flow Designer provides the application logic for automatic classification and email notification. The current solution does not use a separate external API, separate cloud database, or machine-learning model.

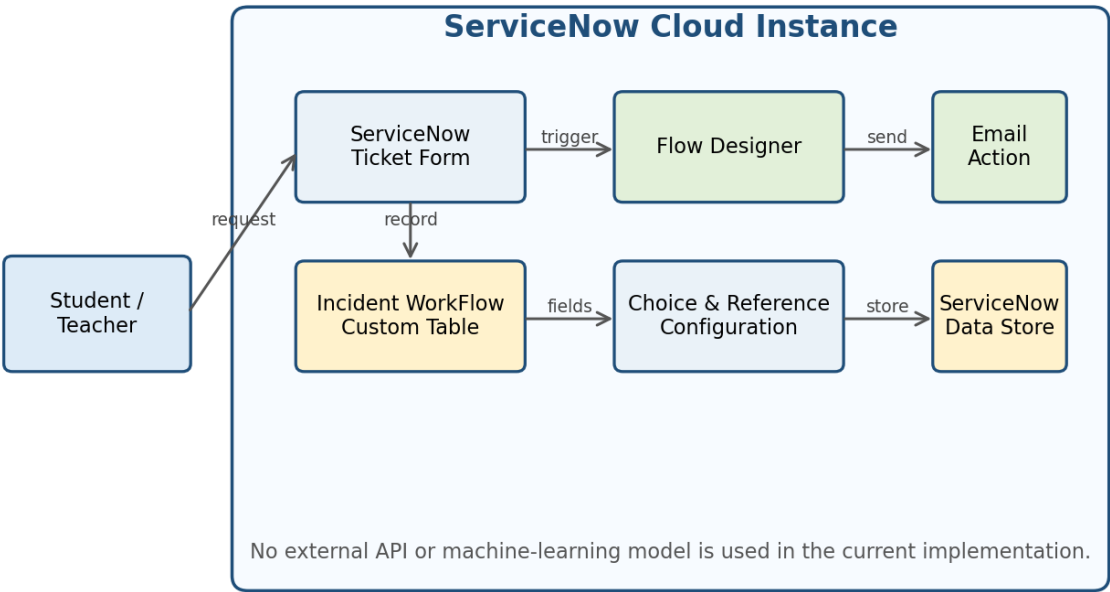


Figure 1. Technical architecture of the current implementation.

2. Components & Technologies

S.No	Component	Description	Technology
1	User Interface	ServiceNow ticket form used by	ServiceNow UI / Forms

		students/teachers and support users.	
2	Application Logic-1	Trigger and keyword-based ticket classification.	ServiceNow Flow Designer
3	Application Logic-2	Dependent Category/Subcategory behavior.	ServiceNow Choice / Dictionary Configuration
4	Application Logic-3	Email confirmation after ticket processing.	Flow Designer – Send Email action
5	Database	Stores ticket records and configured fields such as caller, category, subcategory and state.	ServiceNow platform database / Custom Table
6	Cloud Database	No separate cloud database is used by the project.	Not applicable – data remains within ServiceNow
7	File Storage	No application file-storage service is required for the current workflow.	Not applicable
8	External API-1	No external API is used in the current implementation.	Not applicable
9	External API-2	No external API is	Not applicable

		used in the current implementation.	
10	Machine Learning Model	No machine-learning model is used; classification is rule-based.	Not applicable
11	Infrastructure (Server / Cloud)	The solution runs on the ServiceNow cloud platform / project instance.	ServiceNow Cloud Instance

3. Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is required for the current ServiceNow configuration.	Not applicable
2	Security Implementations	The application uses	ServiceNow access controls / roles

		ServiceNow's platform access and authorization mechanisms. Ticket data should only be accessible to authorized users according to the instance configuration.	
3	Scalable Architecture	The solution is configuration-based. New issue categories and conditions can be added	ServiceNow Tables, Choice Fields, Flow Designer

		to the existing table and Flow Designer logic as requirements grow.	
4	Availability	The application is hosted within the ServiceNow environment. Availability is therefore dependent on the ServiceNow instance and platform.	ServiceNow Cloud
5	Performance	The classification	Flow Designer / ServiceNow platform

		logic uses simple keyword conditions and record updates rather than a separate model or heavy processing. No request-per-second benchmark was measured for this academic project.	
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4. Why These Technologies Were Used

- ServiceNow provides the environment in which the ticket table, form, choices and user references can be configured.
- Flow Designer allows the classification workflow to be implemented without custom scripting.

- Choice and dependent-field configuration helps keep Category and Subcategory values consistent.
- The built-in email action allows the caller to receive a confirmation without a separate messaging service.
- Update Sets provide a way to capture the configuration changes made during the project.

References

C4 model architecture guidance: <https://c4model.com/>

IBM architecture reference supplied in the template: <https://www.ibm.com/cloud/architecture>

AWS architecture reference supplied in the template: <https://aws.amazon.com/architecture>